Grade 6 Name____________________
Unit 3 Assessment Date________________
No Calculator Period_________

Complete the table by rewriting each value in its equivalent forms.

	Fraction Form	Decimal Form	Percent Form
1.	$9\frac{3}{8}$	9.375	937.5%
2.	$\frac{325}{1000}$	0.325	32.5%
3.	$\frac{73}{100}$	0.73	73%

4. Select all of the equivalent forms of 225%.

- ☐ 0.225
- ☐ 2.25
- ☐ 22.5
- ☐ $\frac{225}{1000}$
- ☐ $\frac{225}{100}$
- ☐ $\frac{9}{40}$
- ☐ $\frac{9}{4}$

5. Select all of the equivalent forms of 32.64.

- ☐ 0.3264%
- ☐ 3264%
- ☐ $\frac{816}{25}$
- ☐ $\frac{3264}{1000}$
- ☐ $\frac{3264}{100}$
- ☐ $32\frac{32}{50}$
- ☐ $32\frac{8}{10}$

6. A bolt of fabric is a large rectangular strip of fabric. Alyssa designs her own fabric. The design is then produced and placed on a bolt that is 91.4 meters (m) long and 1.23 m wide.
a. Determine the area of a bolt of fabric with Alyssa's design.

Area (square meters)	112.422

b. Alyssa uses the fabric to make pillowcases. Each pillowcase uses 1.9 square meters of fabric. Determine the number of pillowcases Alyssa can make using a bolt of her fabric.

Number of Pillow Cases	59

7. The average emissions of CO_2 (in grams) per kilometer traveled by a passenger vehicle in Denmark for 2001 and 2019 is shown.

Year	**Average emissions of CO_2 (in grams) per kilometer**
2001	$174\frac{5}{12}$
2019	$107\frac{1}{4}$

Determine the difference in the average emissions of CO_2, in grams, of 30 passenger vehicles in 2019 and 30 passenger vehicles in 2001.

Year	**Average emissions of CO_2 (in grams) per kilometer for 30 cars**
2001	$5232\frac{1}{2}$
2019	$3217\frac{1}{2}$

Students may also find the difference and then multiply by 30.

$$174\frac{5}{12} - 107\frac{1}{4} = 67\frac{2}{12}$$

$$30 \cdot (67\frac{2}{12}) = 2{,}015$$

Difference	$2{,}015$

8. Apple began selling stock on December 12, 1980. The table shows the opening price of one share of stock for each decade since.

Date	**Opening Price of One Share ($)**
12/12/1980	0.1006
12/12/1990	0.2874
12/12/2000	0.2332
12/13/2010	9.9184
12/11/2020	121.5251

Stock is sold and bought for the same price at any given time. Prices are stated to the closest ten-thousandth as stocks are often sold in large quantities.

a. Determine the cost of buying 1200 shares on 12/12/1980.

Cost of 1200 shares on 12/12/1980	$\$120.72$

b. Determine the cost of buying 1200 shares on 12/13/2010.

Cost of 1200 shares on 12/13/2010	$\$11{,}902.08$

c. What is the difference in the cost of 1200 shares of Apple stock on 12/12/1980 and the cost of 1200 shares of Apple stock on 12/13/2010?

Difference in Cost	$11,781.36

d. Determine the total cost of buying 100 shares of Apple stock on 12/12/1990 and buying 100 shares of Apple stock on 12/12/2000.

Total Cost	$52.06

e. Darnell sold his 200 shares of Apple stock on 12/11/2020. He divided the earnings from the sale amongst his 4 children. Determine the amount of money he gave each child.

$$200(121.5251) = 24305.02$$

$$\frac{24305.02}{4}$$

Amount For Each Child	\$6076.25 or \$6076.26

9. For winter, a farmer ordered $\frac{37}{40}$ of a ton of feed for his cattle. Additionally, he ordered $\frac{1}{8}$ of a ton of feed for his goats. At the end of winter, the farmer determined that $\frac{2}{7}$ of the feed was leftover.

Determine the amount of feed, in tons, that was used.

Work shown may vary.

$$\left(\frac{37}{40} + \frac{1}{8}\right) \times \frac{2}{7}$$

$$\frac{42}{40} \times \frac{2}{7} = \frac{3}{10}$$

$$\frac{42}{40} - \frac{3}{10} = \frac{42}{40} - \frac{12}{40} = \frac{30}{40} = \frac{3}{4}$$

or

$$\left(\frac{37}{40} + \frac{1}{8}\right) \times \frac{5}{7}$$

$$\frac{42}{40} \times \frac{5}{7} = \frac{3}{4}$$

Feed Used (tons)	$\frac{3}{4}$